<u>**LAB 1 NOTES**</u>

<u>Binary Number System</u>

- Contains number in base 2
- There are two digits - 0 and 1

<u>Decimal Number System</u>

- Contains number in base 10
- There are ten digits - 0 to 9

<u>Octal Number System</u>

- Contains number in base 8
- There are eight digits - 0 to 7

<u>Hexadecimal Number System</u>

- Contains symbols in base 16
- There are sixteen symbols - 0 to 9 and A to F (A = 10, B = 11, C = 12, D = 13, E = 14, F = 15)

<u>Conversion from decimal to binary:</u>

- Repeat the following until the quotient is 0:
    - Divide a number by 2. Find the quotient and remainder of the number.
    - The quotient becomes the new number to divide.
    - List the remainders from the last one to the first one (**bottom to top**).
- Example 1:
    - Convert 4 to binary.
        - 4 / 2 = 2 (Remainder = 0)
        - 2 / 2 = 1 (Remainder = 0)
        - 1 / 2 = 0 (Remainder = 1)
    - The binary equivalent of 4 is 100.
- Example 2:
    - Convert 15 to binary.
        - 15 / 2 = 7 (Remainder = 1)
        - 7 / 2 = 3 (Remainder = 1)
        - 3 / 2 = 1 (Remainder = 1)
        - 1 / 2 = 0 (Remainder = 1)
    - The binary equivalent of 4 is 1111.

Conversion from binary to decimal:

- Start from the leftmost digit (Least Significant Bit LSB) to the rightmost digit (Most Significant Bit MSB)
- Calculate the number of digits in the binary number (Eg. 1010, number of digits = 4)
- LSB will get the position 0 and keeps moving to MSB with an increment of 1. Let's name this position p.
- Each digit of the binary equivalent will be multiplied by 2^p.
- Find the sum of the results.
- Example 1:
    - Convert 100 to decimal.
        - There are 3 digits.
        - 1 0 0
p=2 p=1 p=0
        - 1 x 2^2 + 0 x 2^1 + 0 x 2^0 = 4 + 0 + 0 = 4
    - The decimal equivalent of 100 is 4.
- Example 2:
    - Convert 1011 to decimal.
        - There are 4 digits.
        - 1 0 1 1
p=3 p=2 p=1 p=0
        - 1 x 2^3 + 0 x 2^2 + 1 x 2^1 + 1 x 2^0 = 8 + 0 + 2 + 1 = 11
    - The decimal equivalent of 1011 is 11.

Conversion from decimal to hexadecimal:

- Repeat the following until the quotient is 0:
    - Divide a number by 16. Find the quotient and remainder of the number.
    - The quotient becomes the new number to divide.
    - List the remainders from the last one to the first one (**bottom to top**).
- Example 1:
    - Convert 4 to hexadecimal.
        - 4 / 16 = 0 (Remainder = 4)
    - The hex equivalent of 4 is 4.
- Example 2:
    - Convert 1234 to hexadecimal.
        - 1234 / 16 = 77 (Remainder = 2)
        - 77 / 16 = 4 (Remainder = 13)
        - 4 / 16 = 0 (Remainder = 4)
    - The hex equivalent of 1234 is 4**13**2. We know the equivalent of 13 in hex is D. So, the final hex equivalent is 4**D**2.

Conversion from hexadecimal to decimal:

- Start from the leftmost digit (Least Significant Bit LSB) to the rightmost digit (Most Significant Bit MSB)
- Calculate the number of digits in the binary number (Eg. 1234, number of digits = 4)
- LSB will get the position 0 and keeps moving to MSB with an increment of 1. Let's name this position p.
- Each digit of the hex equivalent will be multiplied by 16^p.
- Find the sum of the results.
- Example 1:
    - Convert 4 to decimal.
        - There is 1 digit.
        - 1
p=0
        - 1 x 16^0 = 4
    - The decimal equivalent of 4 is 4.
- Example 2:
    - Convert 4D2 to decimal.
        - There are 3 digits.
4 D 2
p=2 p=1 p=0
        - 4 x 16^2 + D x 16^1 + 2 x 16^0 = 4 x 256 + 0 + 2 + 1 = 11
    - The decimal equivalent of 1011 is 11.

Conversion of binary to hexadecimal:

- Method 1
    - Convert binary to decimal, then decimal to hexadecimal
- Method 2
    - There are only 16 digits (from 0 to 7 and A to F) in the hexadecimal number system, so we can represent any digit of the hexadecimal number system using only 4 bits as follows.

Hex	0	1	2	3	4	5	6	7
Binary	0000	0001	0010	0011	0100	0101	0110	0111

Hex	8	9	A	B	C	D	E	F
Binary	1000	1001	1010	1011	1100	1101	1110	1111

- Steps:
  - Take binary number
  - Divide the binary digits into groups of four (starting from right) for integer part and start from left for fraction part.
  - Convert each group of four binary digits to one hexadecimal digit.
- Example:
  - Convert 1010101101001 to hex.
    - (1010101101001)
    - (1 0101 0110 1001)
    - (0001 0101 0110 1001)
    - (1 5 6 9)
    - (1569)
  - The hex equivalent of 1010101101001 is 1569.

<u>Conversion of hexadecimal to binary:</u>

- Method 1
  - Convert hex to decimal, then decimal to binary
- Method 2
  - There are only 16 digits (from 0 to 7 and A to F) in the hexadecimal number system, so we can represent any digit of the hexadecimal number system using only 4 bits as follows.

Hex	0	1	2	3	4	5	6	7
Binary	0000	0001	0010	0011	0100	0101	0110	0111

Hex	8	9	A	B	C	D	E	F
Binary	1000	1001	1010	1011	1100	1101	1110	1111

  - Steps:
    - Write down the hex number and represent each hex digit by its binary equivalent number from the table above.
    - Use 4 digits and add insignificant leading zeros if the binary number has less than 4 digits. E.g. Write 10 (2 decimal) as 0010 in binary.
    - Then concatenate or string all the digits together.
    - Discard any leading zeros at the left of the binary number.

- Example:
    - Convert 3AB2 to binary.
        - 3 A B 2
        - 0011 1010 1011 0010
    - The binary equivalent of 3AB2 is 0011101010110010.

LAB 2 NOTES

- <u>Unsigned Numbers</u>
    - An unsigned binary number is a number with no apparent sign. It is assumed to be positive, including zero.
    - A register of size n bits can store 2^n different numbers, including zero.
- <u>Signed Numbers</u>
    - A signed binary number is a number that can be either positive or negative.
    - The leftmost bit is reserved for the sign. If the bit value is 0, the number is positive. If the bit value is 1, the number is negative.
- <u>Example:</u>
    - +42 in signed 8-bit notation is 0010 1010
    - -42 in signed 8-bit notation is 1010 1010
- <u>One's Complement</u>
    - The one's complement of a binary number is its complement or inverse.
    - Each 0 and 1 are switched so that the 0's become 1's and the 1's become 0's.
    - Example:
        - Given the 8-bit binary value 01001001, find the one's complement.
        - Answer: 10110110
    - The numbers that begin with a 0 are positive. The numbers that begin with 1 are negative.
    - The range of values for a one's complement number is - (2^(n-1) -1) to +2^(n-1) -1 where n = the number of bits.
- <u>Two's Complement</u>
    - The most common method to represent signed numbers is to use the 2's complement system.
    - The range of values for a two's complement number is -2^(n-1) to +2^(n-1) -1 where n = the number of bits.
    - A positive number is converted into the Two's Complement format in the following way:

        - Convert the number to binary.
        - Add leading zeros to the number so it fills n bits.
        - The leftmost sign will be 0.
    - Example:
        - Convert 510 into an 8 bit, two's complement form:
            - Answer:
            - 510 = 1012
            - Fill the bits with leading zeros: 0000 0101
    - A negative number is converted into the Two's Complement format in the following way:
        - Convert the positive form of the number to binary.
        - Add leading zeros to the number so it fills n bits.
        - The leftmost sign will be 0.
        - Find the one's complement of the number.
        - Add 1 to the least significant bit (ie. Add the one's complement and 1).
    - Example:
        - Convert -5 into an 8 bit, two's complement form:
        - Answer:
            - Convert the positive of-5 to binary: 101
            - Fill the bits with leading zeros: 0000 0101
            - Find the one's complement: 1111 1010
            - Add 1 to the number:
            - 1111 1010
+ 1
____
1111 1011

- <u>Two's Complement to Decimal</u>
    - In order to convert a two's complement binary number to decimal, do the following:
        - When the sign bit is 0 (positive), convert the number to decimal.
            - Example:
                - Convert the signed binary number 0101 to decimal.
                - Answer: 5
        - When the sign bit is 1 (negative):
            - Find the one's complement of the number, + 1
            - Convert the number to decimal.
            - Because the sign bit is 1, the decimal number becomes negative.

- Example:
  - Convert the signed binary number 1010 to decimal.
    - Find the one's complement: 0101
    - Add 1 to the number:
    - 0101
      \+ 1
      ____
    - 0110
    - 0110 = 5
    - The original number had a sign bit of 1 (negative), so the final answer is -5

- <u>Binary Addition (Unsigned)</u>
  - To add unsigned binary numbers, do the following:
    - Add leading zeros as needed to the numbers so they both have the same
    - number of n bits.
    - Line-up the second number below the first one.
    - For each column from right to left, do the following:
    - Add the two numbers. If the numbers are both one's, the result is 0 and carry 1 to the next column on the left.
  - <u>Example:</u>
    - Find the sum of 1000 and 100
      - Answer:
      - Add a leading 0 to 100: 0100
      - Line up 1000 and 0100, then add them:
      - 1000
    - \+ 0100
        - ____
      - 1100
  - <u>Example:</u>
    - Find the sum of 1110 and 101
      - Answer:
      - Add a leading 0 to 101: 0101
      - Line up 1110 and 0101, then add them:
      - 1110
        \+ 0101
        _____
      - 10011

- Binary Addition (Signed)
    - To add signed binary numbers, do the following:
        - Find the two's complements of the numbers.
        - Line-up the second number below the first one.
        - For each column from right to left, do the following:
        - Add the two numbers. If the numbers are both one's, the result is 0
            - and carry 1 to the next column on the left.
        - Check for carry-over flags and overflow in the answer.
    - Carry-over flag occurs when the result of an addition or subtraction calculation exceeds the number of allocated bits.
    - If there is no overflow, discard the extra bit in order to obtain the correct answer.
    - Example:
        - Find the sum of -3 and -5 in 2's complement binary formats.
            - -3 and -5 in 2's complement are 1101 and 1011
            - Add the numbers:
            - 1111 -> carry
            - 1101
              + 1011
              ____
            - 11000
            - Discard the carryover bit at the leftmost position.
            - Find the one's complement of the number + 1:
                - 0111 + 1 = 1000
            - Convert to decimal: 8
            - Because the binary number was negative, the final answer is -8.
    - Overflow
        - Overflow is when the result of an addition or subtraction operation goes into the sign bit.
        - Overflow can be seen in the following way:
            - The two sign bits are the same, but the sign bit in the answer is different.
        - Example:
            - Find the sum of 7 and 3 in 4-bit 2's complement binary formats.
                - Answer:
                - 7 and 3 in 4-bit 2's complement are 0111 and 0011
                - Add the numbers:

- 111 -> carry
- 0111
  + 0011
  ____
- 1010
- The result shows an overflow because the sign bit in the answer is different from the sign bits in the numbers.
- The 2’s complement number of 1010, converted to decimal is its complement + 1:
- 0101
  + 1
  _____
- 0010
- 0010 in decimal is 3.
- The sign bit from the addition of the two original numbers is 1 (negative).
- The final answer is -3.
- The answer is wrong. The sum of 7 and 3 is 10.